

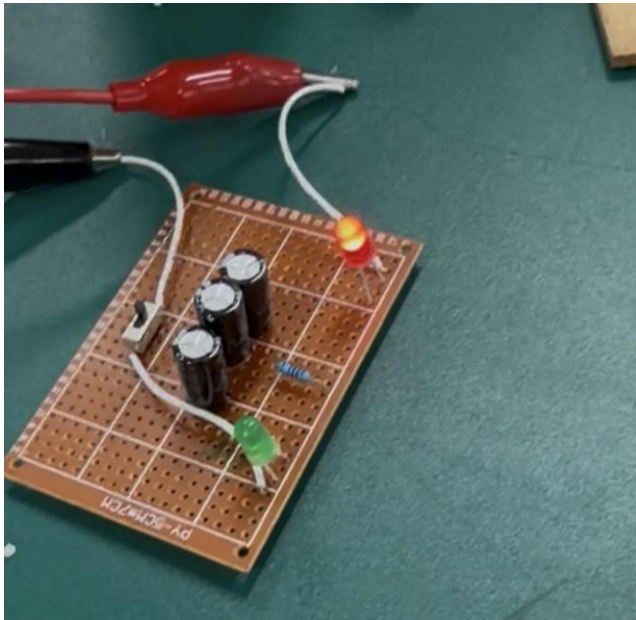
[RC 電路實作與檢測]

組別 15

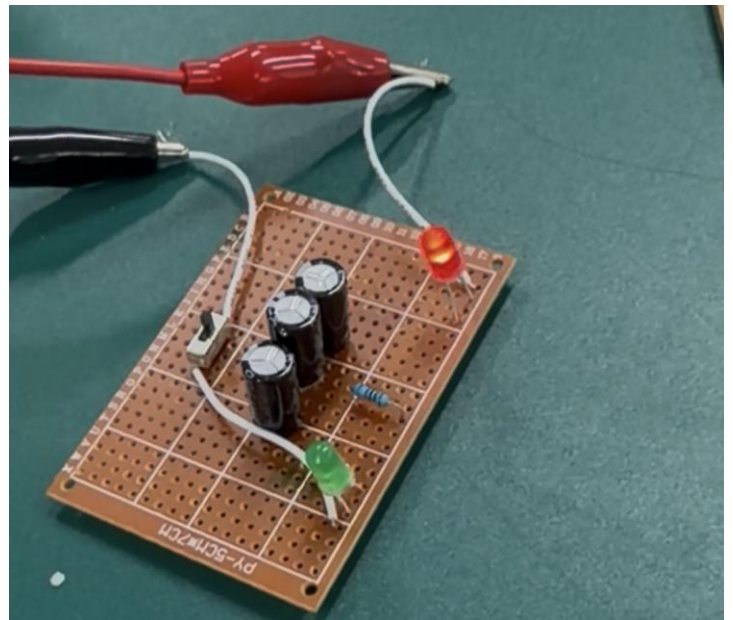
王翊權 E24146107

一、實驗數據

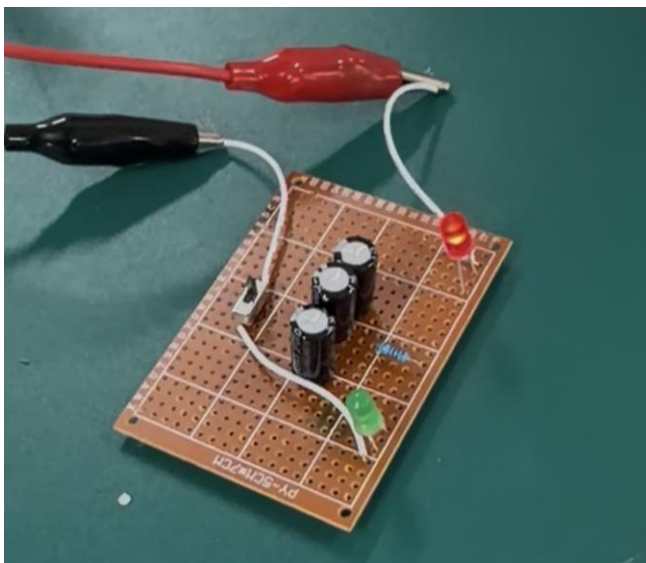
紅LED



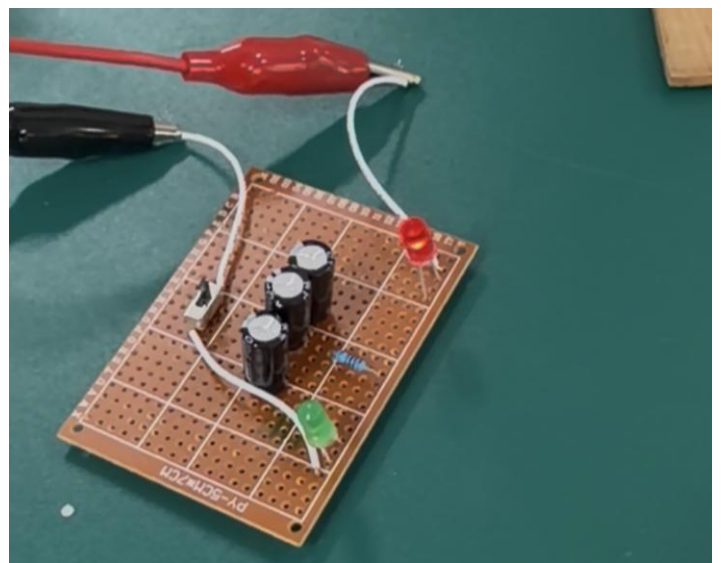
1s



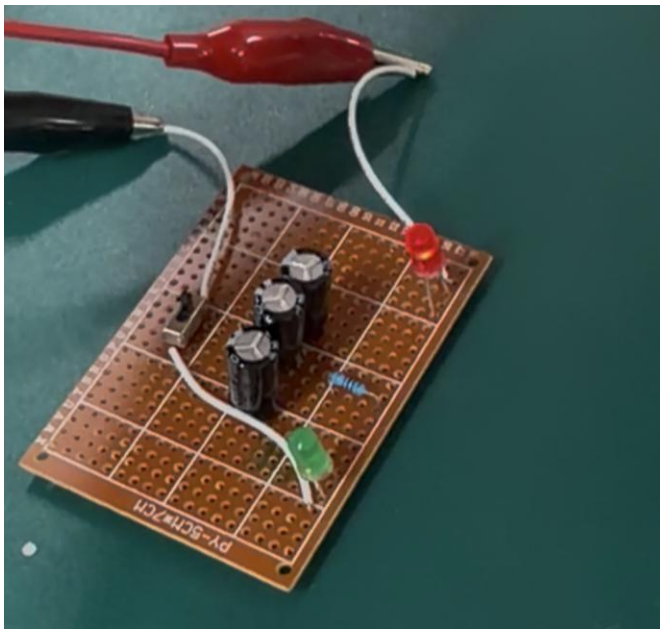
2s



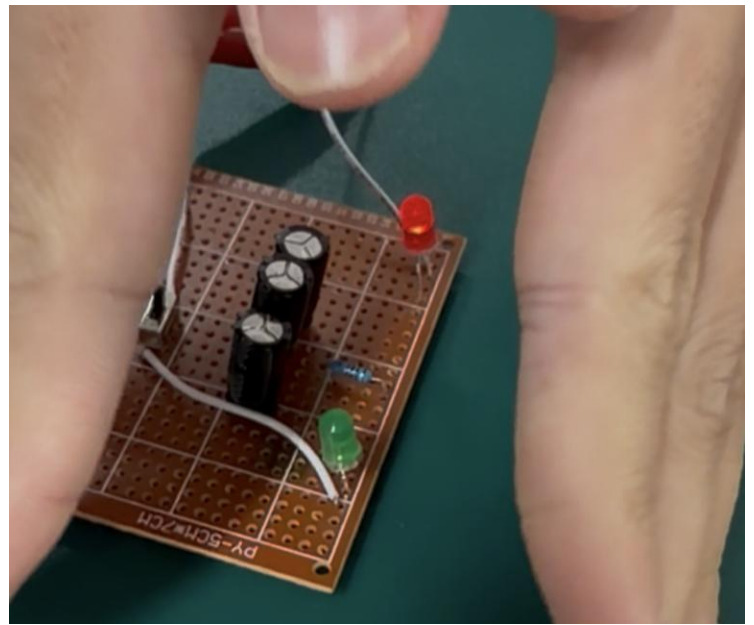
3s



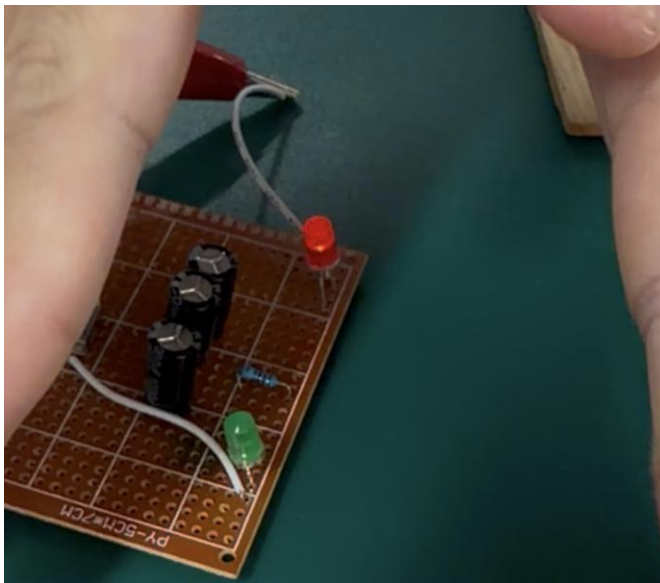
4s



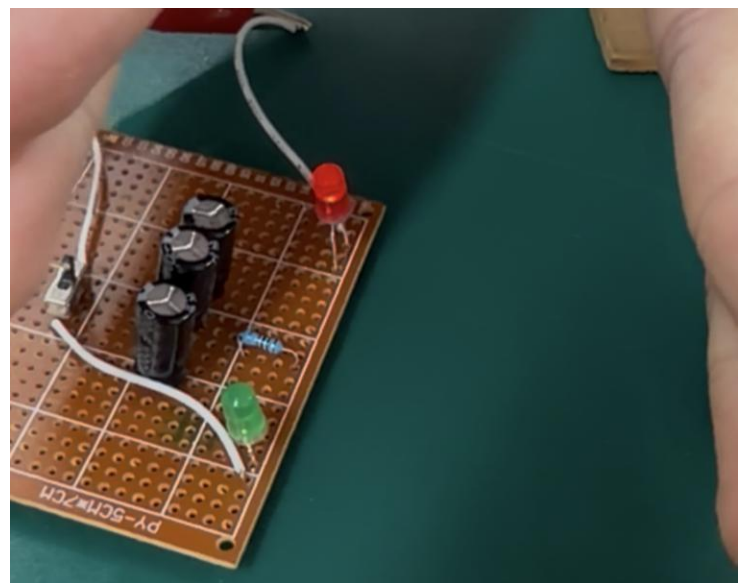
5s



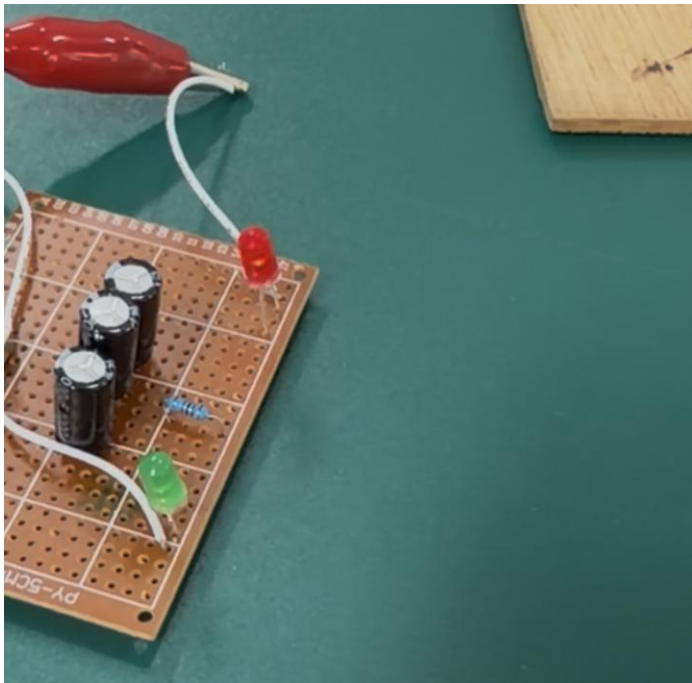
6s



7s

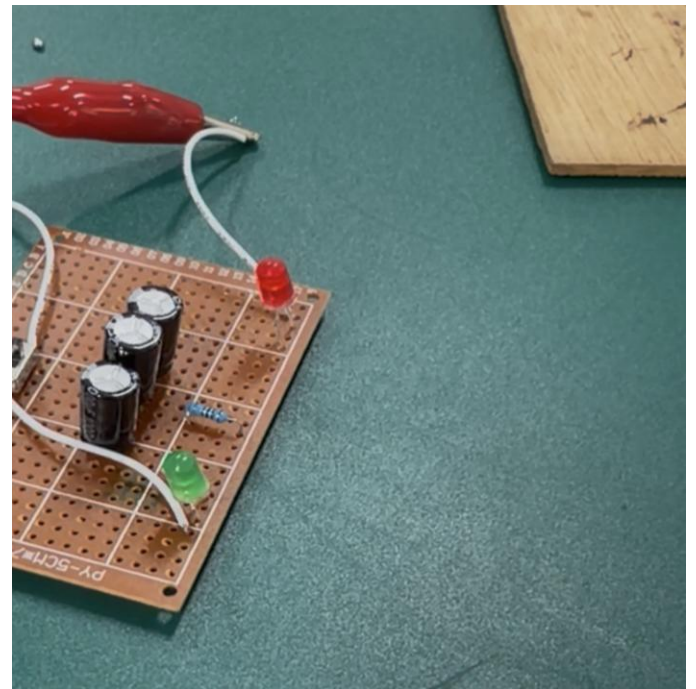


8s

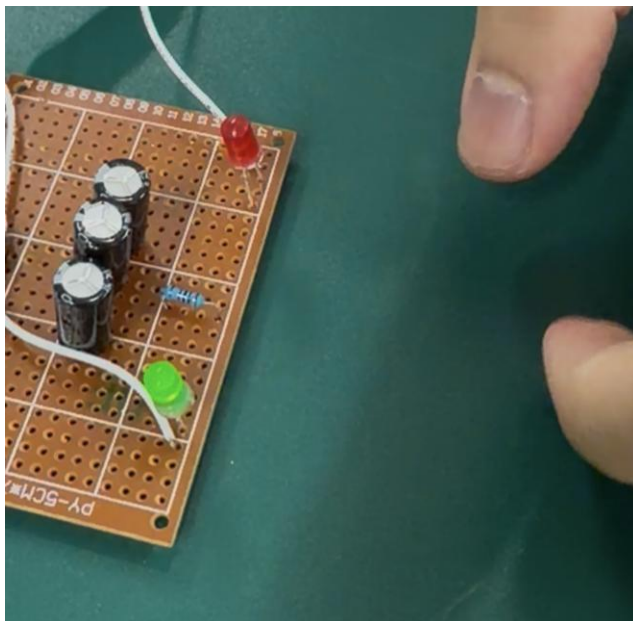


9s

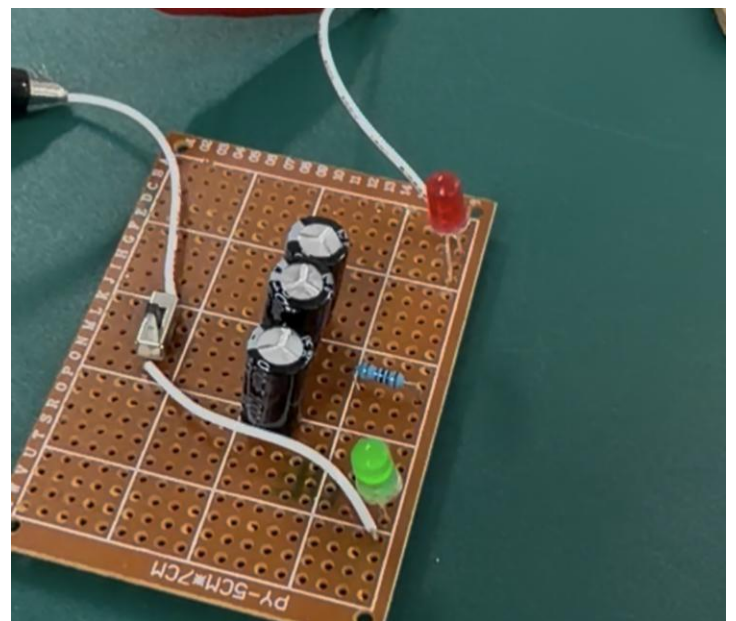
綠 LED



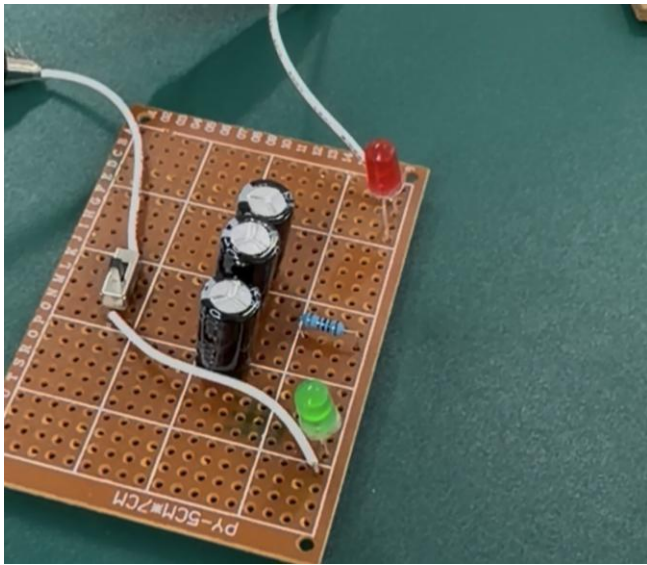
10s



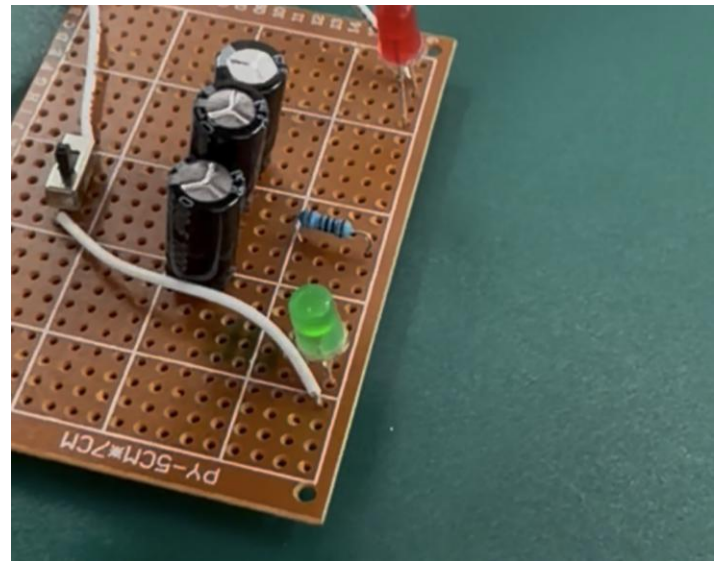
1s



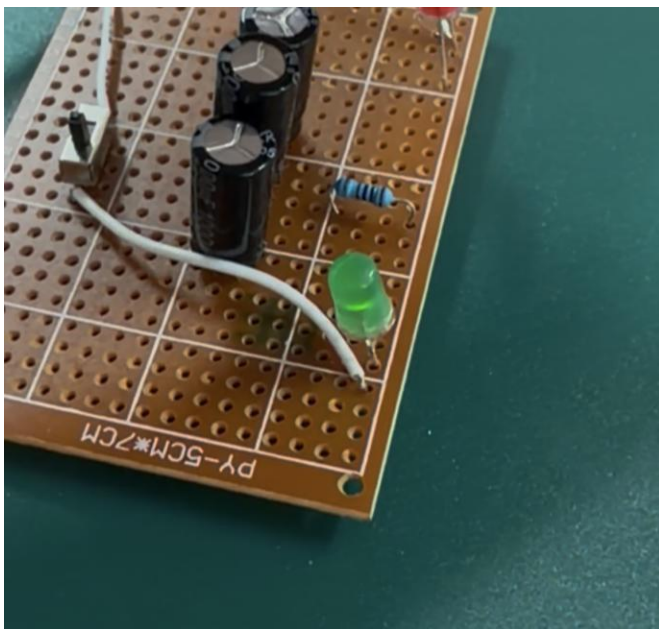
2s



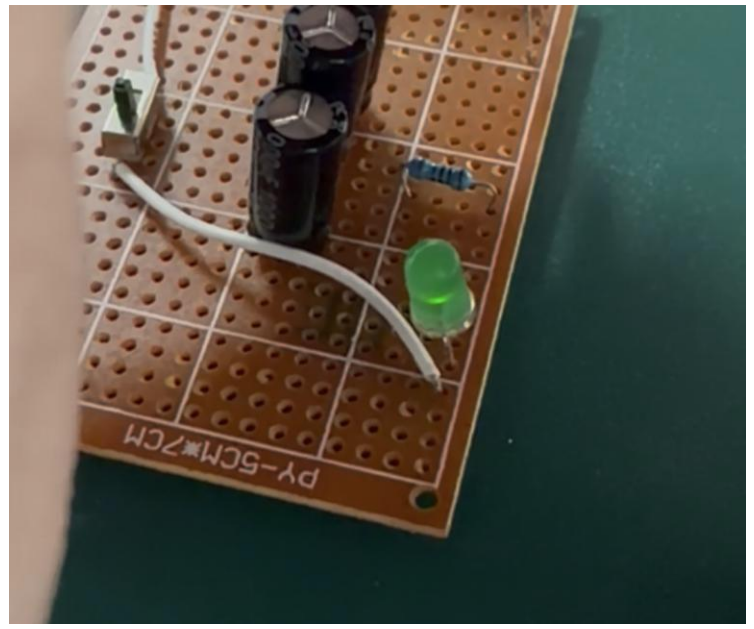
3s



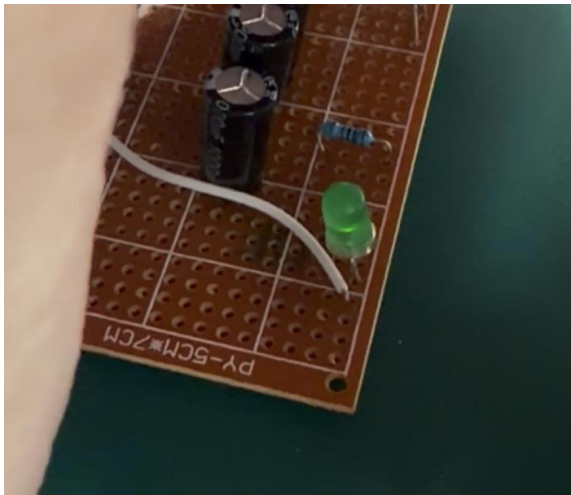
4s



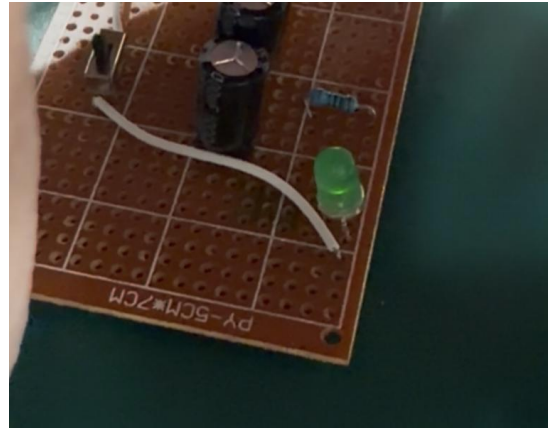
5s



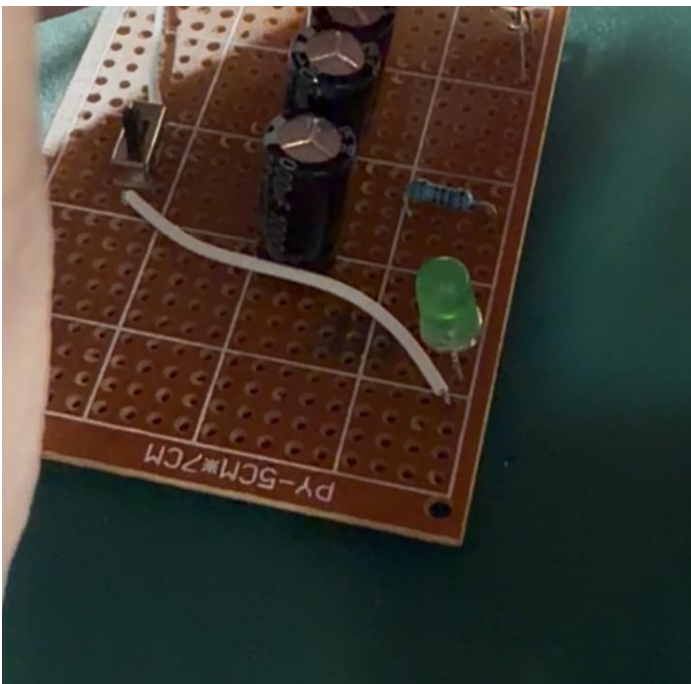
6s



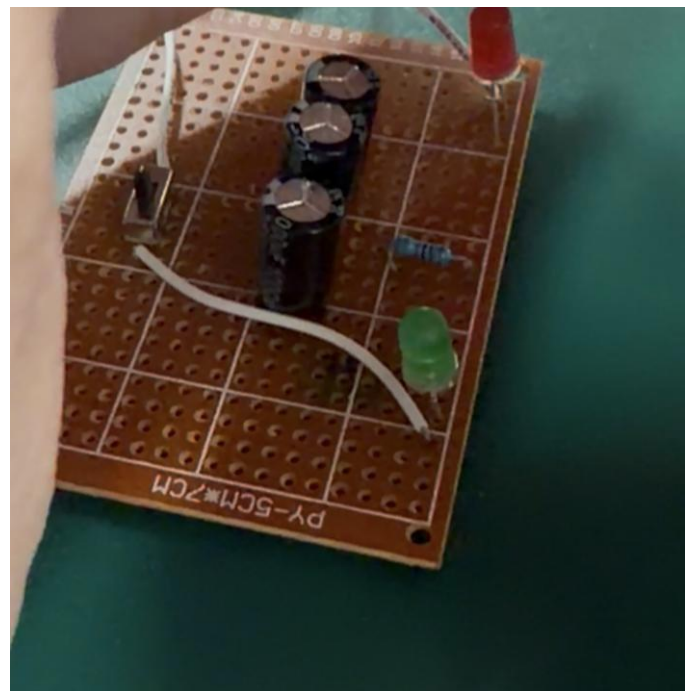
7s



8s

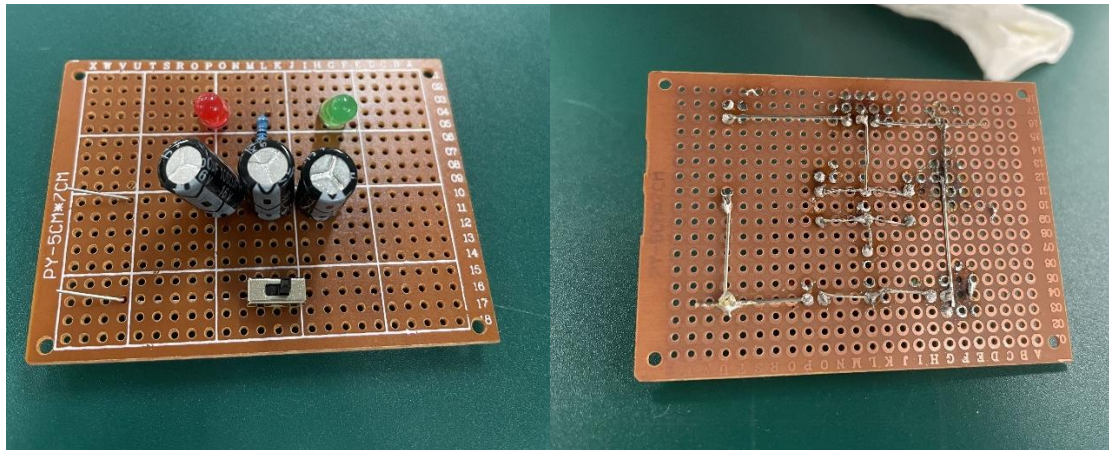


9s



10s

成品圖



二、問題

1. 你能夠想出方法，從 LED 燈亮度隨時間的變化情形估算出 τ 值嗎？

LED 有一個固定的開啟電壓，當電容放電到低於這個電壓時，LED 會突然熄滅。可以拿碼表記錄從「開始放電」到「燈光完全消失」的時間 t 。因為熄滅那一刻的電容電壓就等於 LED 的開啟電壓，我們只要知道電源電壓跟 LED 的開啟電壓，代入公式就可以算出 τ 。

2. 你能利用實驗桌上的儀器，檢驗在前面理論與方法中所說的「在低電流的情況下，LED 電阻值與會與通過它的電流成反比，也就是 LED 在 RC 電路中會產生近似的固定電壓降 a 」這一段敘述嗎？

把 LED 串聯一個電阻後接上電源，改變電源的電，這時候電路裡的電流會明顯變大。此時用電表直接測量 LED 兩端的電壓，如果發現不管電流怎麼變，LED 兩端的電壓都固定在 1.8V 左右，就證明了 LED 在導通狀態下確實具有一個近似固定的電壓降 a 。